INCH-POUND

MIL-DTL-81581/1B
24 July 2006
SUPERSEDING
MIL-H-81581/1A
01 June 1982

DETAIL SPECIFICATION SHEET

HOSE ASSEMBLIES, AIR, LOW PRESSURE, HIGHLY FLEXIBLE

Reactivated after 24 July 2006 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-DTL-81581.

1. SCOPE

1.1 Scope. This specification sheet covers the detail requirements for highly flexible, low pressure, air hose assemblies for use with individual survival equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-81581 - Hose Assemblies, Breathing Oxygen and Air, General Specification for

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 491000B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to thomas.omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 4720

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(Copies of the above specification are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

31C1157	-	Hose Assembly, Air, 7/8 ID; SH3A, S2E, and E2A Aircraft
277AS100	-	Hose Assembly Index, Air, Low Pressure - Highly Flexible

(Copies of the above drawings are available from the Naval Air Technical Data and Engineering Service Command (NATEC), P.O. Box 357031, NASNI, Bldg. 90, San Diego, CA 92135-7031 or https://webjedmics.natec.navy.mil.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-DTL-81581 and as specified herein.

3.2 Qualification. The hoses furnished under this specification shall be a product that is authorized for listing on the applicable qualified products list before contract award (see 4.1.1). Qualification shall be in accordance with groups listed on NAVAIR Index Drawing 277AS100. Qualification approval of any individual hose listed within a group shall automatically qualify a manufacturer for all other items listed within the group provided the conditions set forth in MIL-DTL-81581, 6.3, Qualification, are complied with.

3.3 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.2.

3.4 Talc. Talc shall not be used in the manufacturing process of the hoses furnished under this specification sheet.

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3.5 Design and construction. The hose assembly shall be in accordance with the general specification and with the applicable drawing, NAVAIR 277AS100.

3.6 Performance. Unless otherwise specified herein, the performance of the hose assembly shall be as specified in the general specification, consistent with the applicable performance requirements for the applicable tests and inspections specified herein.

3.6.1 Elongation.

3.6.1.1 Procedure I. When tested as specified in the general specification, the increase in length of the hose shall be not greater than 12 percent of the original hose length.

3.6.1.2 Procedure II. When tested as specified in the general specification, the permanent set of the hose shall be not greater than 2 percent of the original hose length when measured 2 minutes after removal of load.

3.6.2 Strength.

3.6.2.1 Tensile load. When tested as specified in the general specification, the hose shall not show any indications of failure. Two minutes after the load is removed, the permanent set shall be not greater than 0.25 inch.

3.6.2.2. Static load. When tested as specified in the general specification, the hose shall not show a deflection of greater than 25 percent of original value on the outside diameter. With the load removed, the outside diameter of the hose shall be within 5 percent of original value.

3.6.2.3 Weight. When tested as specified in the general specification, the hose shall be not greater than the following applicable weight:

Hose ID (inch)	Weight, maximum
0.500 and 0.625	3 ounces per foot
0.875 and 1.000	4 ounces per foot

3.7 Marking. Included with each length of hose, except hose assembly 31C1157, shall be a separate package of 18 white labels marked for identification in accordance with MIL-STD-130. The year and quarter of cure shall also be included. The label material shall be pressure sensitive vinyl cloth tape with a write-on surface. Each label shall be at least 3/4 inch in width and of sufficient length to completely encircle the applicable hose and overlap at least 0.5 inch.

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4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with the general specification and as specified herein.

4.1.1 Qualification inspection. The qualification inspection of the hose shall consist of the following examinations and tests from the general specification:

Visual examination
Elongation, procedure I
Delamination, procedure I
Odor, procedure I
Cleanliness, procedures I and II
Leakage, procedure I
Flexibility, procedure I (procedure 1 (b) under environmental conditions)
Minimum burst pressure, procedure I
Tensile load, procedure I
Static load, procedure I
Low temperature, procedure I
High temperature, procedure I
Flexibility endurance, procedure I
Ozone-resistance, procedure I
Abrasion, procedure I
Weight, procedure I

4.1.2 First article inspection. The first article inspection of the hose shall consist of the applicable examinations and tests specified in the general specification with the applicable procedure as specified herein.

4.1.2.1 First article samples. The first article samples shall consist of four hoses of the part number contracted for.

4.1.3 Conformance inspection. Conformance inspection shall consist of the following tests from the general specification. The sample size and acceptance criteria shall be as specified in the general specification.

Visual examination
Elongation, procedure I
Delamination, procedure I
Odor, procedure I
Cleanliness, procedure I
Leakage, procedure I
Flexibility, procedure I

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5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the general specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The hose covered by this detail specification sheet is intended for use in distributing air in military anti-exposure and pressure suit systems.

6.2 Acquisition requirements. Acquisition requirements should be in accordance with the general specification.

6.3 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Preparing activity:
Navy - AS

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